SAM VOLUME I

The Spherical Accumulation Kernel and the Weak-Field Limit

Chapter III • Source-bound development draft

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Abstract. A retained source account becomes a spatial field only after its source density, receiving operator and normalization have been specified. This chapter develops the spherical accumulation lift $a(r) = 2GM/(c^2 r)$ as the unique isolated exterior solution of a declared three-dimensional Poisson problem. It derives the source flux, extended-body continuation, shell theorem, mechanical readouts and tidal response, and supplies amplitude and radial-power controls. A finite uniform sphere distinguishes nonzero potential from nonzero acceleration, while a sourced static functional identifies the negative binding-energy account. Recalculation of a frozen Earth–Moon–Sun packet recovers the established Newtonian limit. The universal floor, local clock normalization and spatial metric response retain their distinct roles. The microscopic source-to-density map and the relativistic completion remain explicit physical interfaces.

1. From a retained account to a spatial field

A local account and a field around a source answer different questions. Chapter I constructs a finite source coupled to a continuum field and studies an actual phase clock. Chapter II distinguishes a present output from a predictive state, a timed history, and the energy carried by degrees of freedom that a selected receiver leaves unread. A source can retain or transport an account without that account already specifying an acceleration at a distant point.

The next question is spatial: given a localized source, what profile does a distant receiver encounter? The unified Volume I introduces the spherical accumulation lift [1]

$$a(r) = \frac{r_s}{r}, \qquad r_s = \frac{2GM}{c^2}. \tag{3.1}$$

Here M is the mass assigned to the macroscopic source, G is the gravitational coupling, c is the speed used in the physical normalization, and r is the distance from the center in the static Euclidean receiving problem. The source length r_s is conventionally called the Schwarzschild radius. Naming that length does not make the static calculation below a derivation of a black-hole metric.

This chapter expands the mathematical content of that short formula. It derives the exterior shape from the declared Poisson problem, fixes its normalization by source flux, extends the result through a finite source, and obtains potential, acceleration, circular motion and escape from explicitly defined receivers. It also asks which assumptions these recoveries actually test. The inverse-radius solution is unique within the stated boundary problem; that uniqueness does not establish that the substrate must obey that problem.

The distinction is constructive. A microscopic-to-macroscopic theory must explain why its source produces a mass density, why the receiving operator is the Laplacian in the relevant regime, and why the coupling has the declared normalization. By writing that interface down, the chapter gives later source work a precise target. It does not replace a missing source mechanism with the success of a familiar exterior solution. The source-bound derivations and numerical controls are useful even while that interface remains open.

1.1 Notation and regime

We write the total account as

$$A_{\mathrm{tot}}(\mathbf{x}) = A_0 + a(\mathbf{x}), \qquad A_0 = \frac{1}{\pi \mathscr{R}}, \quad \mathscr{R} = 12. \tag{3.2}$$

The lower-case a is the source lift denoted $A(r)$ in the unified manuscript. This notation prevents an accidental insertion of the floor into a source-only receiver. The structural radix $\mathscr{R}$, physical source radius b, and radial coordinate r have different meanings. None is the quadrupole coupling radius used in Chapter I.

The analysis is static and Newtonian at its core. The field equation is linear; mass density is prescribed; test-body backreaction, rotation, pressure contributions, retardation and nonlinear gravitational self-coupling are outside that core. Where a weak-field clock or metric is mentioned, its additional receiving law is stated separately. Physically using the slow-motion formulas requires small source lift and small speeds relative to c. A mathematical solution can exist outside that regime without licensing the same physical approximation there.

2. The source-to-density interface

Chapter II makes the source-to-density requirement concrete. [2] For a positive energy form B and a linear record

$\boldsymbol{r} = L^T \boldsymbol{y}$, Chapter II gives the ambient minimum

$$C_{\mathrm{amb}}(\boldsymbol{r}) = \boldsymbol{r}^T \mathscr{G}^+ \boldsymbol{r}, \qquad \mathscr{G} = L^T B^{-1} L. \tag{3.3}$$

Here $\mathscr{G}$ denotes the receiver Gram matrix, to distinguish it from Newton's constant G. Restriction to physically allowed source states can increase this minimum. Moreover, a positive displacement cost and a signed change from an existing reference need not coincide: the latter retains a cross term with that reference. These results determine what a declared source and receiver retain and pay. They do not, by themselves, locate that energy in physical space or identify it with the mass that enters a gravitational field equation.

For the present chapter, $\rho(\boldsymbol{x})$ is therefore a supplied, nonrelativistic mass density, and $M = \int \rho \, d^3x$ is its total mass. A microscopic completion would have to identify the contributing physical energy, its units, its spatial incidence, its reference, and the approximation under which its energy density divided by c^2 supplies this ρ. Applying E/c^2 to an arbitrary diagnostic minimum would not establish that completion. Nor does choosing energy units determine the dimensionless structural floor A_0.

This is an explicit interface in the development. We retain the finite-source results without claiming that they already derive the continuum source law. Given ρ, a static three-dimensional Poisson account and its boundary conditions define the receiving field. The local source equation and potential convention are

$$\nabla^2 a = -\frac{8\pi G}{c^2}\rho, \qquad \Phi = -\frac{c^2}{2}a. \tag{3.4}$$

Together they recover $\nabla^2\Phi = 4\pi G\rho$. This normalization imports the measured Newtonian coupling. The inverse-radius shape follows from the stated spatial operator and source symmetry; its physical amplitude requires this coupling and the mass assignment. Keeping those steps separate shows precisely what the spherical calculation proves.

For a slow test body with equal inertial and passive gravitational mass, the Newtonian Lagrangian $L = m|\dot{\mathbf{x}}|^2/2 - m\Phi$ gives

$$\mathbf{g} = -\nabla\Phi = \frac{c^2}{2}\nabla a. \tag{3.5}$$

The test mass cancels from this equation. Its universality is inherited from the specified test-body coupling; it is not proved by the scalar profile alone. For a positive spherical mass, a decreases outward, so this convention gives an inward acceleration.

3. The spherical kernel and its mechanical receivers

3.1 The Green representation and superposition

For a bounded, compactly supported density, the isolated receiving law has the representation

$$a(\mathbf{x}) = \frac{2G}{c^2}\int_{\mathbb{R}^3} \frac{\rho(\mathbf{y})}{|\mathbf{x}-\mathbf{y}|}\, d^3y. \tag{3.6}$$

The kernel singularity is locally integrable in three dimensions. Applying its distributional Laplacian yields the prescribed source equation. This is a dimensionless lift: $G\rho\, d^3y/(c^2|\mathbf{x}-\mathbf{y}|)$ has no units. Nonnegative density produces nonnegative lift under the zero-at-infinity reference.

Linearity implies that a sum of prescribed densities produces the sum of their lifts. This statement applies to the static receiving problem; it is not a law of unrestricted nonlinear relativistic superposition. Two regular solutions with the same density and the same decaying boundary data differ by a globally harmonic decaying function and hence agree, by the harmonic uniqueness theorem. An exterior problem alone needs its inner boundary data or flux and symmetry, as the next calculation makes explicit.

3.2 Exterior uniqueness and its required boundary data

For a spherical vacuum exterior, $\rho = 0$ and

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{da}{dr}\right) = 0, \qquad a(r) = C_0 + \frac{C_1}{r}. \tag{3.7}$$

The condition $a \to 0$ at infinity fixes $C_0 = 0$ but does not fix C_1. Integrating the source equation over a ball enclosing mass M yields

$$4\pi r^2 a'(r) = -\frac{8\pi GM}{c^2}, \qquad a(r) = \frac{2GM}{c^2 r}, \qquad \mathbf{g} = -\frac{GM}{r^2}\hat{\mathbf{r}}. \tag{3.8}$$

Thus spherical symmetry, vacuum, decay, and flux jointly establish the unique exterior kernel within this receiving problem. Decay alone leaves the normalization undetermined. Without decay, an additive constant remains. For a point source the distributional identity is $\nabla^2(1/r) = -4\pi\delta^{(3)}(\mathbf{x})$; the vacuum equation is valid only at $r > 0$. A nonzero $1/r$ term is not regular at the origin and cannot describe a regular vacuum ball containing its center.

More generally, for any enclosing surface S with outward normal,

$$M_{\mathrm{enc}} = -\frac{c^2}{8\pi G}\oint_S \nabla a \cdot d\mathbf{S}. \tag{3.9}$$

In spherical symmetry this reduces to $M(<r) = -c^2 r^2 a'(r)/(2G)$. This formula provides a normalization and sign check independent of the additive potential reference.

3.3 Extended sources and the shell theorem

For a nonnegative, bounded spherical density with compact support, the source law and the regular central condition imply

$$M(<r) = 4\pi\int_0^r \rho(s)s^2\, ds, \qquad a'(r) = -\frac{2GM(<r)}{c^2 r^2}. \tag{3.10}$$

Choosing $a(\infty) = 0$ gives

$$a(r) = \frac{2G}{c^2}\left[\frac{M(<r)}{r} + 4\pi\int_r^\infty \rho(s)s\, ds\right]. \tag{3.11}$$

Differentiating verifies the cancellation of the two local density terms: $(M/r)' = 4\pi\rho(r)r - M/r^2$, while the second term has

derivative $-4\pi\rho(r)r$. Exterior spherical shells therefore contribute to the potential but not to the local acceleration. At $r=0$, bounded central density supplies a regular limit; do not interpret $M(<r)/r$ separately there. Mere finite total mass is weaker: a central cusp proportional to r^{-2} has finite mass near the origin but a divergent central potential. The displayed formula still applies at $r>0$ for such a cusp when the corresponding integrals exist.

For an infinitesimally thin shell of radius b and mass M_s,

$$a_s(r)=\begin{cases}2GM_s/(c^2b), & r<b,\\ 2GM_s/(c^2r), & r>b.\end{cases} \tag{3.12}$$

The potential is continuous at the shell. Its derivative jumps by $-2GM_s/(c^2b^2)$, as required by the surface mass. The field directly on an ideal zero-thickness surface needs a limiting or regularization convention; the formulas here refer to the two sides. Vanishing interior acceleration does not mean vanishing interior potential relative to infinity.

For a uniform solid sphere of mass M and radius b,

$$\rho=\frac{3M}{4\pi b^3}, \qquad a(r)=\begin{cases}\dfrac{GM}{c^2b^3}(3b^2-r^2), & 0\le r\le b,\\ \dfrac{2GM}{c^2r}, & r\ge b,\end{cases} \tag{3.13}$$

$$\mathbf{g}(r)=\begin{cases}-GM\mathbf{x}/b^3, & r<b,\\ -GM\hat{\mathbf{r}}/r^2, & r>b.\end{cases} \tag{3.14}$$

Both a and a' match at b, because this density has no delta-function surface layer. The interior Laplacian is $-6GM/(c^2b^3)$, exactly $-8\pi G\rho/c^2$. The second radial derivative can jump at the density boundary without creating a surface delta function. At the center the field is zero while $a(0)=3GM/(c^2b)$ is nonzero.

3.4 Uniform background, zeros, and potential references

If a proposed full quantity is $A=A_0+a$ with spatially uniform A_0, then $\nabla A=\nabla a$. This is a statement about the spatial derivative, not about all possible effects of A_0. It does not establish that A_0 is absent from clock calibration, a metric, boundary data, or time-dependent dynamics. If the force law uses $a=A-A_0$, it must be stated as such.

Three propositions must be kept distinct:

1. A field that is identically constant on an open region has zero spatial gradient throughout that region.

2. A field that vanishes at one point need not have zero gradient there: $f(x)=x$ has $f(0)=0$ and $f'(0)=1$.

3. Zero gradient at a point does not imply zero field there: $f(x)=1+x^2$ has $f'(0)=0$ and $f(0)=1$.

Consequently, a pointwise cancellation, a zero force, and a global cancellation of the field are different claims. A locally constant shell potential is an especially relevant physical example. A truly global additive constant cancels from all potential differences, including an escape-energy calculation when the same shift is made at both endpoints. A shell's interior constant is not such a global shift: outside the shell its potential changes with radius.

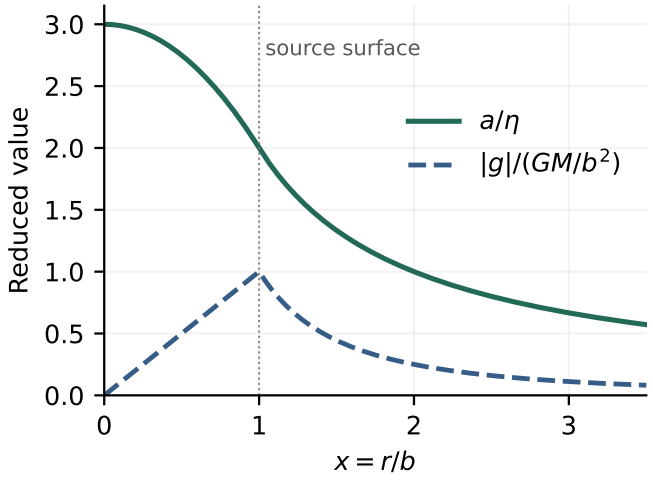


Figure 1: A uniform sphere with $x=r/b$ and compactness $\eta=GM/(c^2b)$. The solid curve is a/η; the dashed curve is acceleration magnitude in units GM/b^2. A finite nonzero central lift coexists with zero central acceleration. These are analytic profiles, not measured data; the physical weak-field regime has $3\eta\ll 1$.

3.5 Circular speed and escape speed test different quantities

For a circular Newtonian test-particle orbit in a static spherical field,

$$v_{\rm circ}^2=r\Phi'(r)=-\frac{c^2r}{2}a'(r). \tag{3.15}$$

For escape to infinity with zero terminal speed and no other barriers, conservative energy balance gives

$$v_{\rm esc}^2=2[\Phi(\infty)-\Phi(r)]=c^2[a(r)-a(\infty)]. \tag{3.16}$$

Only after setting $a(\infty)=0$ may this be written $v_{\rm esc}^2=c^2a$. For an isolated spherical exterior,

$$v_{\rm circ}^2=\frac{GM}{r}, \qquad v_{\rm esc}^2=\frac{2GM}{r}=2v_{\rm circ}^2. \tag{3.17}$$

Inside the uniform sphere,

$$v_{\rm circ}^2=\frac{GMr^2}{b^3}, \qquad v_{\rm esc}^2=\frac{GM}{b^3}(3b^2-r^2). \tag{3.18}$$

The factor-of-two relation is therefore an exterior point-mass result, not a universal spherical relation. At $r=0$ the circular-speed formula is a limiting value, not a finite-radius circular orbit. Adding a concentric exterior shell leaves interior circular dynamics unchanged but increases the squared escape speed by $2GM_s/b$.

3.6 Controls that expose a fitted normalization or wrong power

A rescaled exterior $a_q = q\,2GM/(c^2 r)$ is harmonic for every constant q, but its flux mass is qM, its acceleration is $-qGM/r^2$, and its squared circular and escape speeds are multiplied by q. The vacuum Laplacian check alone therefore cannot validate the amplitude. Treating M as adjustable can conceal this error; an independent source mass and coupling are required for a normalization test.

For a power-law trial $a_p = Kr^{-p}$ at $r > 0$,

$$\begin{aligned} \nabla^2 a_p &= p(p-1)Kr^{-p-2}, \\ g_r &= -\frac{c^2 pK}{2} r^{-p-1}, \\ M_{\rm flux}(r) &= \frac{c^2 pK}{2G} r^{1-p}. \end{aligned} \tag{3.19}$$

The coefficient K has units of length to power p; comparing numerical amplitudes across different p without a reference scale is invalid. One can instead match a dimensionless value a_* at a fixed radius r_* with $a_p = a_*(r_*/r)^p$. For a nontrivial decaying exterior, vacuum and constant flux mass select $p = 1$. The other harmonic radial power, $p = 0$, is constant and carries zero mass flux. For $p > 0$, a positive K gives attraction, but attraction alone does not establish the correct power or source law. When the infinity potential is zero, $v_{\rm esc}^2/v_{\rm circ}^2 = 2/p$ wherever the circular orbit formula applies. Wrong powers can thus alter both radial scaling and this ratio. These are conditional diagnostics, not measurements.

3.7 Calibration at one radius does not validate the radial law

For a clean comparison let the target inverse-radius field be $a_1 = a_*(r_*/r)$, with $a_* > 0$. A wrong-power model calibrated to the potential at r_* is

$$a_p^{\rm pot} = a_* \left(\frac{r_*}{r}\right)^p. \tag{3.20}$$

It agrees with $a_1(r_*)$, but its force magnitude at that same radius is p times the target. Conversely, for $p > 0$, the force-matched model

$$a_p^{\rm force} = \frac{a_*}{p} \left(\frac{r_*}{r}\right)^p \tag{3.21}$$

has the correct acceleration at r_* but only $1/p$ of the target potential there, assuming both potentials vanish at infinity. Its force ratio to the target at any other radius is $(r_*/r)^{p-1}$. Potential and force matching at one radius together select $p = 1$ within this simple family, but only if the potential reference is independently fixed. An arbitrary additive constant can disguise a local potential mismatch while leaving the incorrect radial force scaling unchanged. Measurements or checks at several radii are therefore more discriminating than a single calibrated datum.

3.8 A differential check using the tidal tensor

A useful check beyond the acceleration itself is the Cartesian Jacobian $T_{ij} = \partial g_i/\partial x_j = (c^2/2)\partial_i\partial_j a$. For the spherical exterior, with $n_i = x_i/r$,

$$\partial_i\partial_j a = \frac{2GM}{c^2 r^3}(3n_i n_j - \delta_{ij}), \qquad T_{ij} = \frac{GM}{r^3}(3n_i n_j - \delta_{ij}). \tag{3.22}$$

Its eigenvalues are $2GM/r^3$ in the radial direction and $-GM/r^3$ in each transverse direction. For a small separation $\boldsymbol{\xi}$, $\delta\mathbf{g} = T\boldsymbol{\xi}$ to first order: radial separations stretch and transverse separations compress. The zero trace is the vacuum divergence check. Within the uniform sphere,

$$T_{ij} = -\frac{GM}{b^3}\delta_{ij}, \qquad \operatorname{tr} T = -\frac{3GM}{b^3} = -4\pi G\rho. \tag{3.23}$$

The tidal tensor can be nonzero where the acceleration vanishes, as at the center of the sphere. Conversely, a strictly uniform acceleration has a zero tidal tensor despite a nonzero first derivative of the potential. These distinctions prevent conflating a zero field, zero gradient, and zero Hessian. The radial tidal component jumps across the sharp density boundary, while the acceleration remains continuous. The tensor here is a Newtonian acceleration derivative, not an independently derived relativistic curvature tensor.

3.9 Limits from nonsphericity and external boundary conditions

A general decaying exterior harmonic function can include multipoles,

$$a(r,\Omega) = \frac{C}{r} + \sum_{\ell\geq 1}\sum_{m=-\ell}^{\ell} \frac{C_{\ell m}}{r^{\ell+1}} Y_{\ell m}(\Omega), \tag{3.24}$$

with a real-valued combination understood. Total mass fixes the monopole flux but not the higher multipoles; spherical symmetry removes those terms. A vacuum subdomain subject to remote masses can also contain regular harmonic contributions, such as a uniform-field linear potential or a trace-free quadratic tidal potential. These terms need not decay within a local model whose boundary is finite. Thus local vacuum plus a mass value is insufficient to infer a unique $1/r$ field without symmetry and appropriate boundary conditions. The exact cancellation of exterior shell forces requires concentric spherical shells, not an arbitrary external mass distribution.

3.10 Reproducible check and dimensionless interpretation

The companion script `kernel_weakfield_checks.py` executes 30 exact symbolic assertions and writes a JSON report. It checks vacuum behavior, normalization, force signs, uniform-sphere matching, the shell derivative jump, speed relations, and deliberately wrong amplitude or power controls. No observed data or fitted astrophysical constants enter the checks. For illustration define $x = r/b$ and $\eta = GM/(c^2 b)$, so that $a = \eta\,\alpha(x)$. Then $\alpha = 3 - x^2$ inside a uniform sphere and $\alpha = 2/x$

outside. The report lists reduced accelerations in units GM/b^2 and squared speeds in units GM/b. These reduced quantities can be order unity while the physical weak field remains small, $\eta \ll 1$ (in particular $a(0) = 3\eta \ll 1$). This normalization avoids misrepresenting an order-unity dimensionless kernel as a weak field. Passing these checks establishes consistency with the stipulated Newtonian closure; it supplies neither observational confirmation nor an independent derivation of that closure from microscopic rules.

4. The sourced static field account

A field equation specifies which spatial profile responds to a source. A variational formulation makes the same statement by balancing a penalty for spatial variation against a source coupling. Both terms are necessary. Minimizing a positive gradient term alone, with zero boundary data, would select the zero field rather than the source's accumulation profile.

For a sufficiently regular, localized density, consider the static functional

$$\mathscr{J}[a;\rho] = \frac{c^4}{32\pi G}\int_{\mathbb{R}^3} |\nabla a|^2\, d^3x - \frac{c^2}{2}\int_{\mathbb{R}^3} \rho a\, d^3x. \tag{3.25}$$

Its terms have energy units. Fix the source and vary a while holding the boundary data fixed. Integration by parts, with the boundary term vanishing, gives

$$\delta\mathscr{J} = \int_{\mathbb{R}^3}\left[-\frac{c^4}{16\pi G}\nabla^2 a - \frac{c^2}{2}\rho\right]\delta a\, d^3x. \tag{3.26}$$

Stationarity for every admissible variation yields Eq. (3.4). In this sense the functional receives the prescribed source and returns its static field. It is a continuum representation of the adopted Poisson law, not a microscopic derivation from the finite quadratic forms in Chapter II. It contains no time-evolution law or propagation-speed result.

The positive sign of its gradient term deserves care. It makes the functional convex for fixed ρ and fixed boundary conditions; it does not make Newtonian gravitational binding energy positive. On a regular stationary solution, integration by parts also gives

$$\int |\nabla a|^2\, d^3x = \frac{8\pi G}{c^2}\int \rho a\, d^3x. \tag{3.27}$$

Consequently,

$$\mathscr{J}[a_{\rm sol};\rho] = -\frac{c^2}{4}\int \rho a_{\rm sol}\, d^3x \tag{3.28}$$

$$= \frac{1}{2}\int \rho\Phi\, d^3x \equiv U_{\rm grav}, \tag{3.29}$$

$$U_{\rm grav} = -\frac{c^4}{32\pi G}\int |\nabla a_{\rm sol}|^2\, d^3x. \tag{3.30}$$

For the uniform sphere already constructed, direct radial integration gives

$$U_{\rm grav} = -\frac{3GM^2}{5b}. \tag{3.31}$$

The exterior gradient contribution is included in this result: restricting the field integral to the material interior would omit part of the binding-energy account. The companion interface checker verifies equality of the source and whole-space gradient integrals.

The factor $1/2$ avoids counting each source pair twice. The positive gradient contribution inside $\mathscr{J}$ and the negative on-solution binding energy have different roles. Neither expression licenses identifying a positive microscopic record cost with gravitational self-energy.

These integrals also have a domain. An ideal point source gives a valid exterior kernel and a distributional Poisson equation, but its unregularized self-energy diverges at the origin. The finite-energy calculation applies to suitable extended sources, or to an explicitly stated regularization. It should not be evaluated at the point singularity as though the exterior solution had supplied an interior model.

5. The floor and the choice of local normalization

The structural floor and the local source lift coexist in $A_{\rm total} = A_0 + a$. Their different roles become visible when an instrument's operation is stated. A spatially constant floor has zero gradient, so the acceleration readout is insensitive to it:

$$\boldsymbol{g} = \frac{c^2}{2}\nabla A_{\rm total} = \frac{c^2}{2}\nabla a. \tag{3.32}$$

An ordinary endpoint subtraction also removes the same constant. These are algebraic cancellations. A nonlinear function of the total account requires a separate calculation.

Suppose, as a diagnostic, that one inserts the total account into a raw lapse,

$$N_{\rm raw}(a) = \sqrt{1 - A_0 - a}, \qquad 1 - A_0 - a > 0. \tag{3.33}$$

Normalizing this lapse to its source-free background gives

$$\frac{N_{\rm raw}(a)}{N_{\rm raw}(0)} = \sqrt{1 - \frac{a}{1 - A_0}}. \tag{3.34}$$

Likewise, the ratio between two endpoints is

$$\frac{N_{\rm raw}(a_2)}{N_{\rm raw}(a_1)} = \sqrt{\frac{1 - a_2/(1-A_0)}{1 - a_1/(1-A_0)}}. \tag{3.35}$$

The common multiplicative background factor cancels, but the lift is rescaled. To first order, the endpoint rate ratio differs from unity by $(a_1 - a_2)/[2(1-A_0)]$. It is not generally $(a_1 - a_2)/2$. For $A_0 = 1/(12\pi)$, the residual first-order factor is $1/(1-A_0) \simeq 1.027248616$.

The source-normalized local prescription instead declares its input to be the lift itself:

$$N_{\rm loc}(a) = \sqrt{1 - a}, \qquad \frac{N_{\rm loc}(a_2)}{N_{\rm loc}(a_1)} = \sqrt{\frac{1 - a_2}{1 - a_1}}. \tag{3.36}$$

This convention is the one used for the local endpoint account in the Unified Synthesis. Chapter II's updated discussion makes

explicit that it is a different input from the raw total-account lapse. The two prescriptions can be made formally identical by redefining the lift as $\widetilde{a} = a/(1-A_0)$, but that redefinition must also be carried through the field normalization and every subsequent readout. One cannot retain $a = 2GM/(c^2 r)$ unchanged in both prescriptions and call the difference an automatic cancellation.

Thus the floor remains part of the structural account, while the local convention must be declared at the point where a measurement is defined. Its absence from the source-only static functional is a choice of the local background reference. This chapter does not assign a cosmological stress, energy density, or dynamics to A_0 from that choice.

6. A frozen weak-field packet and what it checks

The recovered CR004 packet provides declared Earth, Moon and Sun gravitational parameters $\mu = GM$, source radii and broad surface-acceleration and escape-speed ranges. Its candidate family changes kernel amplitude or radial power while retaining the receiver definitions. The original result and runner are preserved as sources; no original runner or native campaign was executed for this chapter. We recomputed the standard arithmetic independently from the frozen input file. [3]

For the Earth input, for example,

$$\begin{aligned} \mu &= 3.986004418 \times 10^{14}\ \mathrm{m^3\,s^{-2}}, \\ b &= 6.371000 \times 10^{6}\ \mathrm{m}, \\ c &= 299792458\ \mathrm{m\,s^{-1}}. \end{aligned} \tag{3.37}$$

These give $r_s = 8.870056078 \times 10^{-3}\,\mathrm{m}$ and $a(b) = 1.392254917 \times 10^{-9}$. The source length is millimetric even though the physical body's radius is millions of metres. It is a mass-normalization length, not the size of the object. Applying the two mechanical receivers gives

$$\begin{aligned} |\mathbf{g}(b)| &= 9.820250487\ \mathrm{m\,s^{-2}}, \\ v_{\mathrm{esc}}(b) &= 11186.13569\ \mathrm{m\,s^{-1}}. \end{aligned} \tag{3.38}$$

The other two cases are included in Table 1. The decimal digits expose the arithmetic; they are not a claim of measurement precision. These are spherical idealizations using the packet's selected radii, with no correction for rotation, latitude, topography or atmospheric effects.

Each recomputed acceleration and escape speed falls within the packet's declared range. More importantly for interpreting this result, the same μ and radius are used on both sides of the formula comparison. The packet tests consistent amplitude, radial dependence and receiver use. It does not independently derive G, determine the masses from the finite ledger, or distinguish SAM from Newtonian gravitation in this limit. Agreement is an intended recovery of that limit.

That scope also separates different kinds of evidence. The source result reports its own test verdict; the present chapter checks its displayed arithmetic; the continuum proofs establish consequences of the declared receiving law. Those three statements should not be combined into a single count of independent physical confirmations.

Table 1: Independent arithmetic from frozen CR004 inputs. Escape speeds are in $\mathrm{km\,s^{-1}}$. These values recover the same Newtonian formulas used by the source packet; they are not a new observational fit.

Source	$a(b)$	$\lvert\mathbf{g}\rvert$ ($\mathrm{m\,s^{-2}}$)	v_{esc}
Earth	1.39225×10^{-9}	9.82025	11.1861
Moon	6.28226×10^{-11}	1.62490	2.37618
Sun	4.24501×10^{-6}	274.200	617.675

The original inverse-square candidate uses $(r_s/r)^2$, so its failure mixes a changed radial power with a large amplitude change. The reference-radius controls in the present derivation isolate that issue more cleanly. They are additional analytical controls, not newly reported CR004 measurements.

7. What the scalar field supplies to later measurements

The same source lift can enter several measurements without making those measurements the same operation. Its value defines the Newtonian potential under the chosen normalization. Its gradient gives a local acceleration. A clock comparison requires a rule for proper time at each endpoint, while a signal delay requires a rule along an entire path. The latter two rules require more physical structure than a scalar Poisson solution alone supplies.

The distinction can be stated within a conventional weak-field metric description. For a static, approximately isotropic geometry, let

$$ds^2 = -[1 - a + O(a^2)]c^2dt^2 + [1 + \gamma a + O(a^2)]d\boldsymbol{x}^2. \tag{3.39}$$

Here t is normalized to the source-free local reference, and γ describes the leading spatial response. This is additional metric input in the standard post-Newtonian description. [7] The temporal term yields the leading stationary-clock rate $d\tau/dt = 1 - a/2 + O(a^2)$ and, in the slow-motion limit, the acceleration $\boldsymbol{g} = (c^2/2)\nabla a$. It does not determine γ.

For a null path in these coordinates, the same metric instead gives

$$c\,dt = \left[1 + \frac{1+\gamma}{2}a + O(a^2)\right] d\ell. \tag{3.40}$$

The leading photon-road coefficient is therefore $(1+\gamma)/2$, whereas the stationary endpoint clock has coefficient $1/2$. The familiar road coefficient of unity requires the additional choice $\gamma = 1$, as in general relativity. A lapse alone cannot establish that choice or the associated light-deflection and delay predictions. Endpoint conversion of coordinate time into an observer's proper time also remains a separate step.

This is the appropriate stopping point for the present chapter. The scalar source problem establishes a spatial kernel, its normalization, and its Newtonian consequences under stated assumptions. Chapter IV can then develop endpoint clocks with their reference and motion terms. Chapter V can develop photon roads with their spatial metric, path, and endpoint conventions. Reusing a across those chapters makes the source account common; each receiving operation still needs its own justification. In particular, the weak-field construction does not establish an exact strong-field metric, a horizon, or the physical domain of the square-root lapse by extrapolation.

8. What remains to derive from the source

The chapter's exact mathematical endpoint is a conditional field construction. Given a specified static density, the Poisson receiver and the isolated boundary condition, the accumulation lift is fixed. The uniform-sphere example demonstrates that this construction handles source interiors without substituting the exterior singularity at the center. The shell example demonstrates that a locally vanishing force does not imply a vanishing infinity-referenced potential account. The wrong-kernel controls show why a matched value at one radius is not the same as a matched field law.

Four further results would strengthen the physical chain.

1. **A source-to-density construction.** Specify how the original allowed source states, their reference energy and their physical localization produce $\rho(\mathbf{x})$. A nonnegative selected record cost is not automatically the complete gravitating mass.

2. **A continuum-limit operator.** Derive the Laplacian, its isotropy and its domain from the relevant source evolution or state explicitly why it is the intended macroscopic receiving law. A finite-state closure theorem is not by itself a spatial continuum limit.

3. **A coupling and mass-calibration rule.** Fix the conversion to physical mass and the relation between source coupling and measured G without fitting every body or receiver independently. Binding energy and additional field contributions must be counted under a common reference.

4. **A dynamical and metric extension.** Identify the causal evolution law and the clock, spatial and light-propagation receivers. Static inverse-square recovery does not determine their nonlinear completion.

These are suitable computational targets with separate stopping conditions: a defined source map; a controlled continuum approximation with an error bound; one shared calibration; and a complete receiving law whose different observables can be checked without changing its parameters. A failure at one of these interfaces should remain visible rather than being absorbed into a new normalization.

9. Conclusion

The spherical kernel organizes a familiar weak-field limit in a single dimensionless source lift. Its $1/r$ shape follows from the spherical vacuum Poisson equation, its coefficient follows from the specified mass flux, and its interior continuation follows from the source density. Potential samples that lift, acceleration differentiates it, and mechanical escape compares its value with a declared destination. The operations are connected, but they are not interchangeable.

The floor remains part of the upstream account. A uniform term disappears under a gradient, while a nonlinear endpoint normalization requires its own explicit convention. Similarly, mechanical recovery fixes the temporal potential at leading order only after a clock law is supplied; it does not fix spatial curvature or the complete photon road.

Chapter IV can therefore begin with a well-defined source field and ask how endpoint clocks receive it. The present chapter leaves that next operation clear, while retaining the source-to-field construction as an explicit physical task.

Sources, evidence and reproducibility

The companion archive includes the editable manuscript, figure source, exact symbolic checker, frozen-input arithmetic, source snapshots and content hashes. The 42 symbolic checks (30 kernel checks and 12 interface checks) and 12 frozen-input checks were executed for this draft. They are consistency checks, not independent empirical confirmations. No original SAM runner, GEN6 campaign or physical experiment was rerun.

The Courtroom branch distinguishes its scoped weak-field result from GPS, Shapiro and strong-field closure. Its G224 summary describes earlier continuum and discrete routes, but the retrieved public G224 index has no original source URL or verdict. The original derivation was not recovered here. LC03 is an imported-evidence mechanism replay, rather than a new solution of a microscopic source-to-field evolution. These historical records motivate the interface; the explicitly displayed derivation supplies this chapter's conditional mathematical claim. [4, 5, 6]

References

[1] S. Brady, OpenAI ChatGPT and Codex, *SAM Unified Technical Synthesis—Volume I: Substrate and Accumulation*, 1 September 2026, Sec. 2 and Appendix A.4. Owner-supplied manuscript.

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[3] SAM Research Project, *CR004: Weak-field A-kernel external contact*. Frozen premises, result, precommit and runner. Result and source directory. Snapshots retrieved 16 October 2026 UTC.

[4] SAM Research Project, *A-kernel / weak-field branch: scope and readout contracts*. Branch source.

[5] SAM Research Project, *G224 archive index*, SAMA. Public index. Index inspected; original derivation not recovered.

[6] SAM Research Project, *LC03 mechanism replay*. Result. Imported-source replay scope retained.

[7] C. M. Will, *The Confrontation between General Relativity and Experiment*, Living Reviews in Relativity **17**, 4 (2014). doi:10.12942/lrr-2014-4. Used for the conventional post-Newtonian metric comparison, not current experimental bounds.